DRUGS · SPECIAL ENT AGENTS

Botulinum toxin

Botulinum toxin blocks the release of acetylcholine at nerve terminals, weakening muscle and stopping cholinergic glands from secreting, for about 3 to 4 months. ENT surgeons inject it for spasmodic dysphonia, sialorrhoea and Frey's syndrome, cricopharyngeal dysfunction, palatal and other myoclonus, vocal tremor, laryngeal granulomas, facial synkinesis and hyperkinesis after facial palsy, first bite syndrome, and facial aesthetics. Its doses are in units that are specific to each product and not interchangeable.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

BOTULINUM TOXIN

	SUMMARY
Preparations	OnabotulinumtoxinA (Botox), abobotulinumtoxinA (Dysport), incobotulinumtoxinA (Xeomin); botulinum toxin B (rimabotulinumtoxinB) for resistant cases
How it works	Cleaves SNARE proteins (SNAP-25 for type A) in the presynaptic terminal, preventing acetylcholine release; recovery requires new terminal sprouting (3 to 6 months)
ENT uses	Adductor spasmodic dysphonia (thyroarytenoid injection); abductor spasmodic dysphonia (posterior cricoarytenoid); essential vocal tremor; sialorrhoea (parotid and submandibular glands under ultrasound); Frey's syndrome (intradermal); cricopharyngeal achalasia (dysfunction); laryngeal contact granulomas (rest the folds); synkinesis and hyperkinesis after facial palsy; palatal myoclonus causing objective tinnitus; first bite syndrome; facial aesthetics

	SUMMARY
Typical doses	Adductor spasmodic dysphonia commonly 0.625 to 2.5 units of onabotulinumtoxinA per vocal fold through an EMG-guided needle; sialorrhoea in adults incobotulinumtoxinA 100 units in total (30 units into each parotid and 20 units into each submandibular gland); Frey's syndrome about 1 to 2.5 units per square centimetre of the area mapped by a starch-iodine (Minor) test
Onset and duration	Effect from 1 to 3 days, peak at 1 to 2 weeks, lasting about 3 to 4 months
Avoid	Myasthenia gravis and Lambert-Eaton syndrome; infection at the site; pregnancy and breastfeeding; caution with aminoglycosides (potentiation)

CHAPTER TWO

2 Using it in ENT

Practice.

2.1 Spasmodic dysphonia

A focal laryngeal dystonia: strained, strangled, effortful speech with voice breaks (adductor type) or breathy breaks (abductor type), better with laughing, singing or whispering. Injections into the thyroarytenoid-lateral cricoarytenoid complex through the cricothyroid membrane under EMG guidance, or by a transoral or transnasal approach, restore fluency; patients expect a few days to a couple of weeks of breathiness and mild aspiration of thin fluids after each injection.

2.2 Sialorrhoea

Injected into the parotid and submandibular glands under ultrasound, it reduces drooling in neurological disease (Parkinson's, cerebral palsy, motor neurone disease) for several months; dysphagia can worsen if the toxin spreads to the pharyngeal muscles.

2.3 Frey's syndrome

Gustatory sweating and flushing over the parotid region after parotidectomy (aberrant regeneration of parasympathetic fibres into sweat glands): map the area with the Minor starch-iodine test and inject intradermally; benefit lasts many months.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Local weakness beyond the target (breathy voice and aspiration after laryngeal injection; dysphagia after salivary or cricopharyngeal injection; facial asymmetry and ptosis after facial injection), dry mouth, flu-like symptoms, antibody-mediated resistance after frequent high doses, and very rarely distant spread with generalised weakness.

3.2 Interactions

Aminoglycosides and other drugs that affect neuromuscular transmission enhance its effect.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - Mechanism (SNAP-25 cleavage) and duration.

- Technique and side effects of injection for **spasmodic dysphonia**.
- **Frey's syndrome** and the Minor test.
- Units are **product-specific**.

4.2 Common pitfalls

COMMON PITFALLS

- Converting doses between products as if units were equal.
- Not warning a patient about breathiness and aspiration after laryngeal injection.

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